

Name _____ Date _____ Period _____

Domain and Range: All Three Notations

Algebra II | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

Vocabulary

Domain — the set of all _____ values, the _____-values.

Range — the set of all _____ values, the _____-values.

Set notation — uses _____ with the bar | , read "such that."

Interval notation — uses _____ if an endpoint is included, _____ if it is excluded.

Continuous — the graph is _____, with no breaks or gaps.

Discrete — the graph is made of _____ points.

Part 1 — From a Set of Ordered Pairs

Least to greatest, and each value written only once.

Example 1. $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$

Domain: { _____ }

Range: { _____ }

You Try 1. $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$

Domain: { _____ }

Range: { _____ }

Part 2 — From a Table

Example 2.

x	-3	-1	0	2	5
f(x)	9	1	0	4	25

Domain: { _____ } Range: { _____ }

You Try 2.

x	-2	0	1	4
f(x)	7	7	-3	10

Domain: { _____ } Range: { _____ }

Part 3 — The Three Notations

The first row is done for you. Fill in the rest.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{x \mid x \in \mathbb{R}\}$	$(-\infty, \infty)$
all reals greater than or equal to 0	$x \geq 0$	_____	_____

all reals less than 5	$x < 5$	_____	_____
all reals from -3 to 3, including both	$-3 \leq x \leq 3$	_____	_____
just the numbers -2, 0, and 4	(none)	_____	_____

Key idea: bracket [] = endpoint _____. Parenthesis () = endpoint _____. ∞ always takes a _____.

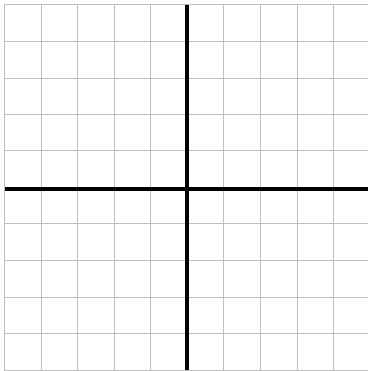
You Try 3. all reals greater than -8 $\rightarrow$ inequality _____ interval _____

You Try 4. $y \leq 12$ $\rightarrow$ in words _____ interval _____

Part 4 — From a Graph

Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.

A. Discrete. Plot (-4, 2), (-2, -3), (1, 4), (3, -1).

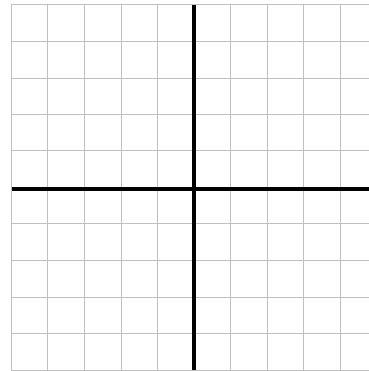


Domain (set): _____

Range (set): _____

Continuous or discrete? _____

B. Continuous. Graph $y = x + 1$ from $x = -4$ to $x = 3$.



Domain (inequality): _____

Domain (interval): _____

Range (interval): _____

Part 5 — Continuous or Discrete?

Write continuous or discrete.

- The number of students in each class period _____
- The outside temperature over one full day _____
- The total cost of n concert tickets at \$45 each _____

Exit Ticket

Given { (1, 5), (3, 5), (4, -2) } write the domain and range in set notation, then say whether interval notation could be used.

Domain _____ Range _____

Could interval notation be used? _____

Domain and Range from Problem Situations

Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

Vocabulary

Independent variable — the _____, the one you choose. Goes on the ____-axis.

Dependent variable — the _____, it depends on the input. Goes on the ____-axis.

Function domain — every input the _____ will accept.

Contextual domain — every input the _____ actually allows. Usually _____.

Part 1 — Name the Variables

Which quantity is the input and which is the output?

1. Hours you work and the money you earn

independent: _____ dependent: _____

2. Number of tickets bought and the total cost

independent: _____ dependent: _____

3. Seconds since a ball was thrown and its height

independent: _____ dependent: _____

Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is $V(x) = x^3$.

What domain does the **equation** x^3 accept? _____

What domain does the **box** allow? _____

Why? _____

Big idea: the equation may accept numbers the _____ does not. Always ask what makes sense.

Part 3 — Domain and Range in Context

Give the contextual domain and range. Use an inequality and interval notation.

Example 1. Concert tickets cost \$12 each. You may buy at most 8. Total cost $C(t) = 12t$.

Domain: _____ Discrete or continuous? _____

Range: _____

You Try 1. A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.

Domain: _____ interval _____ Discrete or continuous? _____

Range: _____ interval _____

You Try 2. A parking garage charges \$3 per hour and is open 6 a.m. to 10 p.m. (16 hours).

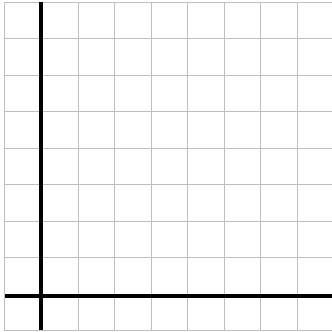
Domain: _____ Range: _____

Part 4 — Reading a Situation from a Graph

Each square is 1 unit. Plot the points, then answer in context.

A water tank is drained. Gallons g after m minutes:

m	0	1	2	3	4
g	8	6	4	2	0



Answer in context.

Domain: _____

Range: _____

What does the domain mean here?

What does a range of 0 mean?

Part 5 — Does the Situation Restrict the Domain?

Write yes or no, and if yes, give the restriction.

- The side length of a square garden _____
- The temperature in a freezer over one day _____
- The number of buses needed for a field trip _____

Practice

Give the contextual domain and range for each. Show your thinking on the lines.

- A phone plan costs \$30 per month plus \$5 per gigabyte, up to 20 GB.

Domain _____ Range _____

- A theater has 250 seats. Tickets are \$9 each, and t tickets are sold.

Domain _____ Range _____

- A diver descends from the surface to 40 meters below sea level.

Range of depth _____ Why negative? _____

Exit Ticket

A taxi charges a \$4 flat fee plus \$2 per mile, for trips up to 10 miles. Cost $C(m) = 4 + 2m$.

Domain _____ Range _____

Why does the range start at 4, not 0? _____

Name _____ Date _____ Period _____

Continuous vs. Discrete

Algebra II | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can decide whether a domain and range are continuous or discrete, and write each one using the notation that fits.

Vocabulary

Continuous — every value between two points is included. The graph is an _____ curve or line.

Discrete — only certain separated values count. The graph is _____.

Whole numbers — _____ **Integers** — _____

Part 1 — Sort the Situations

Write C for continuous or D for discrete.

Situation	C or D	Why?
The number of chairs in a classroom	_____	_____
The distance a car travels	_____	_____
Your age measured in years and days	_____	_____
The number of pets a family owns	_____	_____
The weight of a watermelon	_____	_____

Part 2 — The Same Function, Two Ways

A cube-shaped box has side x inches, so its volume is $V(x) = x^3$.

x	1	2	3	4
$V(x) = x^3$	1	8	27	64

Case A: the box can be any size at all.

Domain _____ Range _____

Case B: the box is built out of 1-inch cube blocks, so x must be a whole number.

Domain _____ Range _____

Same equation, different _____ → different domain. The context decides.

Part 3 — Which Notation Fits?

Continuous domains use _____ notation, like $[0, 5]$.

Discrete domains get _____ in braces, like $\{0, 1, 2, 3\}$, because there are gaps between the values.

You Try 1. A stadium sells 0 to 5 hot dogs per person. Domain _____

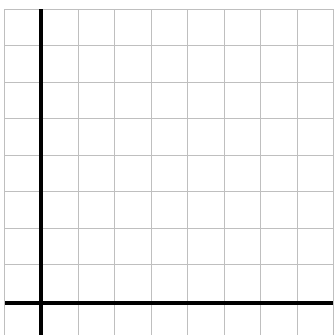
You Try 2. A candle burns from 10 cm down to 0 cm. Domain of heights _____

You Try 3. Why can You Try 1 not use interval notation? _____

Part 4 — Graph Both Kinds

Each square is 1 unit. Plot each, then name the domain and range.

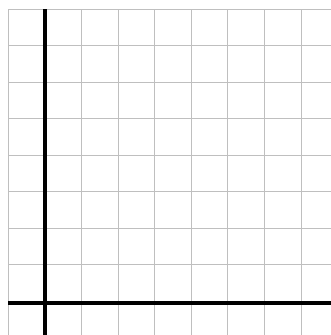
A. Discrete. A tutor charges \$2 per session, up to 4 sessions. Plot (1, 2), (2, 4), (3, 6), (4, 8).



Domain _____

Range _____

B. Continuous. A candle burns 2 cm per hour from 8 cm. Graph $h = 8 - 2t$ for $0 \leq t \leq 4$.



Domain _____

Range _____

Part 5 — Quick Check

1. Should a discrete domain ever be written as an interval? _____
2. Can a continuous range include negative numbers? _____
3. Is money spent at a vending machine continuous? _____

Practice

Decide continuous or discrete, then write the domain in the notation that fits.

1. A runner covers any distance from 0 to 5 miles.

C or D _____ Domain _____ Why? _____

2. A vending machine sells 0 to 6 drinks.

C or D _____ Domain _____ Why? _____

3. Water in a bathtub drops from 30 gallons to 0.

C or D _____ Range _____ Why? _____

4. A class of 24 students is split into equal teams of 4.

C or D _____ Domain (teams) _____

5. Explain in your own words why a discrete domain cannot use interval notation.

Exit Ticket

A school orders pizzas for a party. Each pizza feeds 6 students, and they order up to 5 pizzas.

Domain (pizzas) _____ Range (students fed) _____

Continuous or discrete, and why? _____

Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph the square root and absolute value parent functions from a table and analyze their key attributes: domain, range, intercepts, symmetry, and minimum.

Vocabulary

- Parent function** — the _____ function in a family, with nothing added, subtracted, or multiplied.
- x-intercept** — where the graph crosses the x-axis. There, $y = \underline{\hspace{1cm}}$. **y-intercept** — there, $x = \underline{\hspace{1cm}}$.
- Symmetry** — the graph can be _____ onto itself.
- Minimum** — the _____ y-value the function reaches.

Part 1 — The Square Root Function, $f(x) = \sqrt{x}$

Fill in the table. Watch what happens with negative inputs.

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$						

Why are the first two undefined? _____

Part 2 — The Absolute Value Function, $f(x) = |x|$

Absolute value gives distance from zero, so the output is never negative.

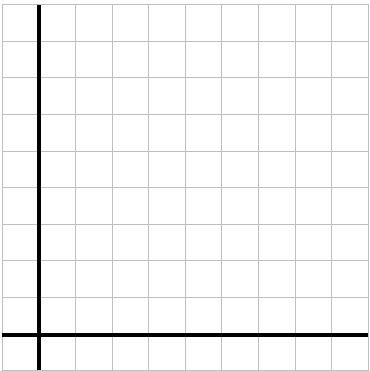
x	-5	-2	-1	0	1	2	5
$f(x) = x $							

What shape does this make? _____

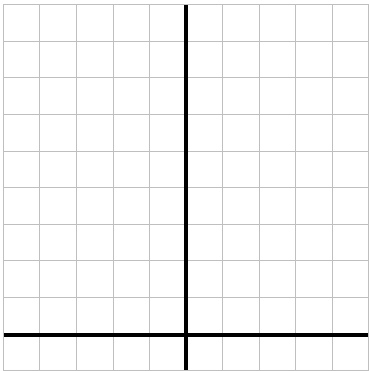
Part 3 — Graph Them Both

Each square is 1 unit. Plot your table values, then connect them smoothly.

A. $f(x) = \sqrt{x}$ (x from 0 to 9)



B. $f(x) = |x|$ (x from -5 to 5)



Which quadrant only? _____

Fold line? _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = \sqrt{x}$	$f(x) = x $
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Domain (interval)	_____	_____
Range (interval)	_____	_____
x- and y-intercept	_____	_____
Symmetry	_____	_____
Minimum	_____	_____
Increasing / decreasing	_____	_____

Biggest difference: their _____. Only _____ refuses negative inputs.

Part 5 — Name That Function

Use only the attributes given. No equation.

- Domain and range both $[0, \infty)$, no symmetry, always increasing. _____
- Domain all reals, range $[0, \infty)$, symmetric across the y-axis, minimum 0. _____

Practice

Evaluate, then answer. Show your work where there is room.

- Evaluate: $\sqrt{25} =$ _____ $\sqrt{0} =$ _____ $|-7| =$ _____ $|0| =$ _____

- Explain why $\sqrt{-9}$ is not a real number.

- Name every x-value where $|x| = 4$. _____ How many? _____

- Name every x-value where $\sqrt{x} = 4$. _____ How many? _____

- Items 3 and 4 have different numbers of answers. What does that tell you about the two graphs?

- On $[1, 4]$, give the minimum and maximum of $f(x) = \sqrt{x}$. min _____ max _____

Exit Ticket

Both functions have a minimum of 0 and pass through the origin. Give one attribute that tells them apart, and explain why it happens.

Attribute: _____

Why: _____

Name _____ Date _____ Period _____

Parent Functions II: x^3 and $\sqrt[3]{x}$

Algebra II | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph the cubic and cube root parent functions from a table and analyze their key attributes, including their rotational symmetry about the origin.

Vocabulary

Rotational symmetry — turning the graph _____ about the origin lands it back on itself.

Reflectional symmetry — the graph can be _____ across a line onto itself.

Cube root — $\sqrt[3]{a}$ asks: what number _____ gives a ? Negatives _____ allowed, unlike square roots.

Part 1 — The Cubic Function, $f(x) = x^3$

Fill in the table. Notice what a negative input produces.

x	-2	-1	0	1	2
$f(x) = x^3$					

A negative input gives a _____ output, because a negative cubed stays negative.

Part 2 — The Cube Root Function, $f(x) = \sqrt[3]{x}$

This undoes the cubic. Compare it to your Part 1 table.

x	-8	-1	0	1	8
$f(x) = \sqrt[3]{x}$					

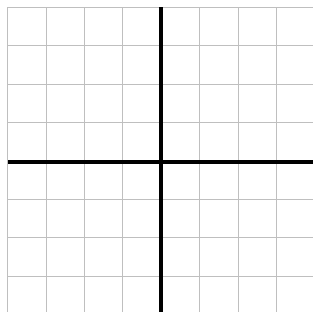
What do you notice next to Part 1? _____

We will come back to that on Day 10 - these two are _____.

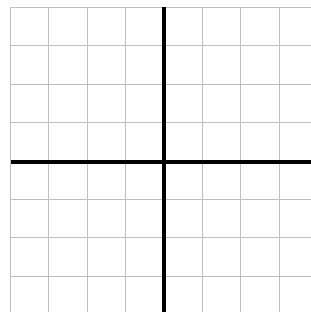
Part 3 — Graph Them Both

Each square is 1 unit. Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$ steep S-curve



B. $f(x) = \sqrt[3]{x}$ flatter S-curve



Passes through the _____

Mirror line between them: _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = x^3$	$f(x) = \sqrt[3]{x}$
Domain (interval)	_____	_____

Range (interval)	_____	_____
x- and y-intercept	_____	_____
Symmetry	_____	_____
Maximum or minimum?	_____	_____
Increasing / decreasing	_____	_____

These two share every attribute above. The difference is the _____ of the curve.

Part 5 — Compare to Day 4

1. Why does $\sqrt[3]{x}$ accept negatives when $\sqrt{x}$ does not? _____
2. Which of the four functions so far have a minimum? _____

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $(-3)^3 =$ _____ $\sqrt[3]{-27} =$ _____ $\sqrt[3]{64} =$ _____ $\sqrt[3]{0} =$ _____
2. Solve $x^3 = 125$. $x =$ _____ How many real answers? _____
3. Solve $x^2 = 25$. $x =$ _____ How many real answers? _____
4. Items 2 and 3 give different counts. Explain why cubing behaves differently from squaring.

5. $f(x) = x^3$ on the interval $[-1, 3]$: minimum _____ maximum _____
6. True or false: every real number has exactly one real cube root. _____
7. Which of the four parent functions so far accept negative inputs?

Exit Ticket

A function has domain and range both $(-\infty, \infty)$, passes through the origin, has 180° rotational symmetry, and no maximum or minimum.

Name two parent functions this could be: _____

What extra information would tell them apart? _____

Name _____ Date _____ Period _____

Parent Functions III: $1/x$ and Asymptotes

Algebra II | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.

Vocabulary

Asymptote — a line the graph gets _____ to but never settles on.

Vertical asymptote — a vertical line the graph _____. It sits where the function is undefined.

Horizontal asymptote — a horizontal line describing end behavior. It _____ be crossed in other functions.

Reciprocal — the reciprocal of a is _____. Zero has _____ reciprocal.

Part 1 — Build the Table for $f(x) = 1/x$

Fill in the outputs. One input has no output at all.

x	-2	-1	-1/2	0	1/2	1	2
$f(x) = 1/x$							

Why is $x = 0$ undefined? _____

Part 2 — What Happens Near Zero and Far Out

Follow the pattern in each list, then describe it.

x	1	0.5	0.1	0.01
$1/x$	1	2	10	100

As x gets close to 0, the outputs grow _____. So there is a vertical asymptote at _____.

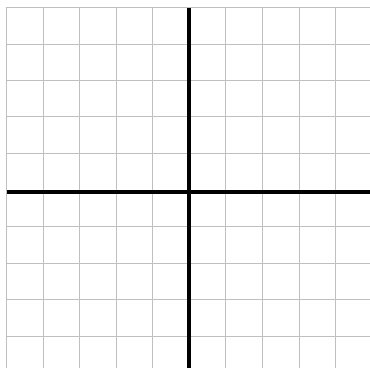
x	1	10	100	1000
$1/x$	1	0.1	0.01	0.001

As x gets very large, the outputs approach _____. So there is a horizontal asymptote at _____.

Part 3 — Graph It

Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y -axis.

$f(x) = 1/x$



Record what you see.

Vertical asymptote: _____

Horizontal asymptote: _____

How many branches? _____

Which quadrants? _____

Any intercepts? _____

Draw both asymptotes as dashed lines.

Part 4 — Key Attributes of $f(x) = 1/x$

Domain: _____ set notation _____

Range: _____ interval _____

Symmetry: reflectional across _____, and rotational _____ about the origin.

Maximum or minimum? _____ Intercepts? _____

This is the first parent function with _____ and the first whose domain has a _____ in it.

Part 5 — True or False

1. A graph may cross a vertical asymptote. _____
2. A graph may cross a horizontal asymptote. _____
3. $f(x) = 1/x$ has a y-intercept. _____
4. The domain of $1/x$ is every real number. _____

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $1/4 =$ _____ $1/(-2) =$ _____ $1/(1/5) =$ _____
2. Solve $1/x = 4$. $x =$ _____ Solve $1/x = -1$. $x =$ _____
3. Is there any x with $1/x = 0$? Explain.

4. As x grows very large and negative, what happens to $1/x$? _____

5. Fill in the pattern.

x	-0.1	-0.01	0.01	0.1
$1/x$				

Approaching 0 from the left the outputs go toward _____, from the right toward _____.

6. Why is $1/x$ the only parent function so far with a gap in its domain?

Exit Ticket

A function has no intercepts, a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, and two branches in Quadrants I and III.

Name it: _____ Domain in set notation: _____

Name _____ Date _____ Period _____

Parent Functions IV: b^x and $\log_b b$

Algebra II | Unit 1 | Day 7 | TEK A2.2.A

Objective: I can graph exponential and logarithmic parent functions for $b = 2, 10$, and e , analyze their key attributes, and notice that their domains and ranges trade places.

Vocabulary

Exponential function — the variable is in the _____, as in $f(x) = 2^x$.

Base — the number being raised to a power. We use $b = 2, 10$, and _____.

Logarithm — log base b of a asks: what _____ on b produces a ?

Natural log — _____ means log base e . **Common log** — _____ means log base 10 .

Part 1 — The Exponential Function, $f(x) = 2^x$

Fill in the table. Negative exponents give fractions, never negative outputs.

x	-2	-1	0	1	2	3
$f(x) = 2^x$						

Can 2^x ever equal 0 or a negative number? _____

So there is a horizontal asymptote at _____ and a y-intercept at _____.

Part 2 — The Logarithmic Function, $f(x) = \log_2(x)$

This undoes the exponential. Compare it to your Part 1 table.

x	1/4	1/2	1	2	4	8
$f(x) = \log_2(x)$						

What happened to the table from Part 1? _____

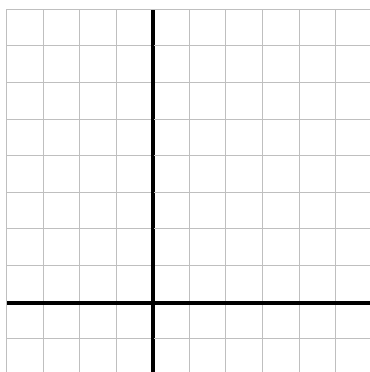
What is $\log_2(0)$? _____

So there is a vertical asymptote at _____ and an x-intercept at _____.

Part 3 — Graph Them Both

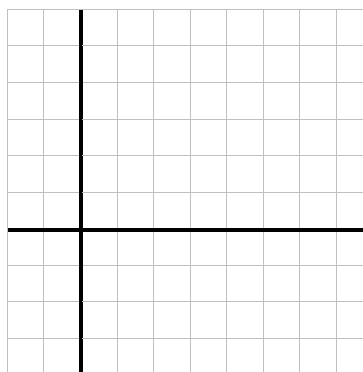
Each square is 1 unit. Sketch the asymptote as a dashed line first, then plot.

A. $f(x) = 2^x$



Asymptote: _____

B. $f(x) = \log_2(x)$



Asymptote: _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = b^x$	$f(x) = \log \text{ base } b$
Domain (interval)	_____	_____
Range (interval)	_____	_____
Intercept	_____	_____
Asymptote	_____	_____
Increasing / decreasing	_____	_____

Look at the first two rows. The domain and range have _____, and so did the intercepts and asymptotes. That is a strong hint these are _____ - Day 10.

Part 5 — Changing the Base

All bases greater than 1 give the same shape. Only the steepness changes.

x	0	1	2
2^x	1	2	4

x	0	1	2
10^x	1	10	100

Which grows faster, and why? _____

What do all three exponential graphs share? _____

Exit Ticket

A function has domain $(0, \infty)$, range $(-\infty, \infty)$, an x-intercept at $(1, 0)$, and a vertical asymptote at $x = 0$.

Name the family: _____ Name its inverse family: _____

Name _____ Date _____ Period _____

Comparing All Seven Parent Functions

Algebra II | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare the key attributes of all seven parent functions, name a function from its attributes alone, and find a relative maximum or minimum on a given interval.

Vocabulary

Relative maximum — the _____ y-value on a given interval.

Relative minimum — the _____ y-value on a given interval.

A function can have no overall max or min but still have one _____.

Part 1 — The Master Comparison Table

Fill this in from your Day 4-7 notes. This is your reference sheet for the rest of the year.

Function	Domain	Range	Symmetry	Asymptotes
$\sqrt{x}$	_____	_____	_____	_____
$ x $	_____	_____	_____	_____
x^3	_____	_____	_____	_____
$\sqrt[3]{x}$	_____	_____	_____	_____
$1/x$	_____	_____	_____	_____
b^x	_____	_____	_____	_____
log base b	_____	_____	_____	_____

Part 2 — Intercepts and Extremes

Function	Intercept(s)	Minimum?	Maximum?
$\sqrt{x}$	_____	_____	_____
$ x $	_____	_____	_____
x^3 and $\sqrt[3]{x}$	_____	_____	_____
$1/x$	_____	_____	_____
b^x	_____	_____	_____
log base b	_____	_____	_____

Only _____ of the seven have a minimum, and _____ has a maximum. Only _____ has no intercept at all.

Part 3 — Name That Function

Use only the attributes listed. Name the function and sketch its shape in the box.

- Domain and range both all reals except 0; two branches; asymptotes at $x = 0$ and $y = 0$.

Function: _____

- Domain all reals, range $[0, \infty)$, folds across the y-axis, minimum of 0.

Function: _____

- Domain $(0, \infty)$, range all reals, passes through $(1, 0)$.

Function: _____

- Domain and range both $[0, \infty)$, no symmetry, no asymptotes.

Function: _____

Part 4 — Relative Max and Min on an Interval

A graph rises and falls between $y = 1.5$ and $y = -1.5$ over the interval from 0° to 360° .

Relative maximum on that interval: _____ Relative minimum: _____

Now consider $f(x) = x^3$ on the interval $[-2, 2]$.

Relative maximum: _____ Relative minimum: _____

x^3 has no overall max or min, but on a _____ it has both. Why? _____

Part 5 — Sort by One Attribute

Which parent functions have a restricted domain? _____

Which have any asymptote? _____

Which have range all real numbers? _____

Practice

Use your comparison tables. Show reasoning where there is room.

1. Two parent functions pass through $(1, 1)$ and $(0, 0)$. Name them. _____

2. Which parent function never touches either axis? _____

3. Which two are reflections of each other across $y = x$? Give two different pairs.

4. A graph has a minimum but no maximum, and no asymptotes. Name both possibilities.

5. $f(x) = |x|$ on the interval $[-3, 1]$: minimum _____ maximum _____

6. Explain why a function can have a relative maximum on an interval but none overall.

Exit Ticket

Two parent functions have domain and range that are both all real numbers, and both have 180° rotational symmetry.

Name them: _____ Name one function with a restricted range:

Name _____ Date _____ Period _____

Modeling with a Parent Function: the Open Box

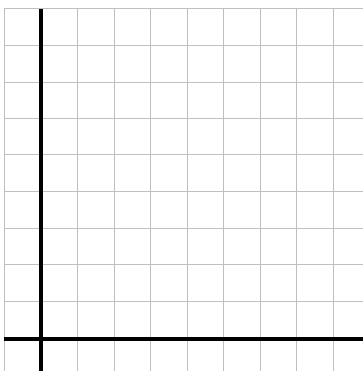
Algebra II | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

Objective: I can build a function that models a real situation, decide which domain makes sense for the context, and use a table or graph to find a maximum.

The Problem

Start with a flat sheet of cardboard **20 inches by 10 inches**. Cut a square of side **x** out of each of the four corners, then fold the flaps up to make an open-top box.

Sketch the sheet with the corner squares marked, and label one side x .



Part 1 — Build the Function

Each cut removes x from BOTH ends of a side, so each side loses $2x$.

Height of the box: _____

Length of the box: _____ Width of the box: _____

Volume $V(x)$ = length $\cdot$ width $\cdot$ height = _____

Multiplied out: $V(x)$ = _____ Which parent function family? _____

Part 2 — Find the Domain That Makes Sense

The equation accepts any real number. The cardboard does not.

Why must x be greater than 0? _____

The short side is 10 inches. What happens if $x = 5$? _____

So the contextual domain is _____ interval _____ Open or closed?

Part 3 — Make a Table

Compute $V(x) = x(20 - 2x)(10 - 2x)$ for each value. Round to the nearest tenth.

x	0.5	1	1.5	2	2.5	3	4	4.5
$V(x)$								

Between which two x -values does the volume stop growing and start shrinking? _____

Largest volume in your table: _____ Is that certainly the true maximum?

Part 4 — Refine the Maximum

Test values between 2 and 2.5 to close in.

x	2.0	2.1	2.11	2.2	2.3
$V(x)$					

Maximum volume $\sim$ _____ when $x \sim$ _____

Box dimensions then: _____

Range of the function on this domain: _____ interval _____

Part 5 — Interpret in Context

1. What does $V(1) = 144$ mean in words? _____
2. Why is $x = 5$ excluded but $x = 4.5$ allowed? _____
3. Would a domain of all reals make sense here? _____

Practice — A Second Box

A sheet is 16 in. by 8 in. Squares of side x are cut from the corners. Show your work.

1. Write the volume function.

$V(x) =$ _____

2. Which side limits the domain, and what is the limit? _____
3. Contextual domain: _____
4. Compute $V(1)$ and $V(2)$. Show your arithmetic.

$V(1) =$ _____ $V(2) =$ _____

5. Explain why $V(4) = 0$ even though 4 is a positive number.

Exit Ticket

A sheet is 12 in. by 12 in. Squares of side x are cut from the corners and the sides folded up.

Volume function: _____ Contextual domain: _____

Explain why the upper limit is 6: _____

Inverses from Graphs and Tables

Algebra II | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or a graph, use $f^{-1}(x)$ notation, and recognize that a function and its inverse are reflections across the line $y = x$.

Vocabulary

Inverse — the relation that _____ the original.

Notation — we write _____ for the inverse of $f(x)$. Careful: this does _____ mean $1/f(x)$.

In a table — to find the inverse, _____ in every pair.

On a graph — a function and its inverse are reflections across the line _____.

Part 1 — Inverses from Ordered Pairs

Swap the coordinates in each pair.

Example 1. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{ \text{_____} \}$

Domain of $f = \text{_____}$ which becomes the _____ of f^{-1} .

You Try 1. $g = \{ (1, 5), (2, 8), (3, 11) \}$ $g^{-1} = \{ \text{_____} \}$

Part 2 — Inverses from a Table

Fill in the inverse table by trading the rows.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

Now the inverse:

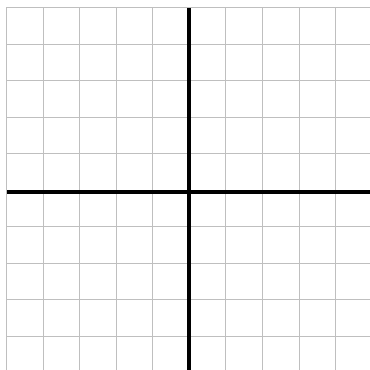
x	-6	-1	4	-7	7
$f^{-1}(x)$					

The two tables are the same information, just _____.

Part 3 — Graph a Function and Its Inverse

Each square is 1 unit. Plot f , then plot f^{-1} , then draw the line $y = x$ through the origin at 45° .

A. Plot $f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$ and its inverse.



B. What do you notice?

The two graphs are mirror images across the line _____

Which points stay put under the swap?

If (a, b) is on f , then on f^{-1} you find _____

$f^{-1} = \{ \text{_____} \}$

Part 4 — Domain and Range Trade Places

For any relation: domain of $f =$ _____ of f^{-1} , and range of $f =$ _____ of f^{-1} .

Function	Domain of f	Range of f	Domain of f^{-1}	Range of f^{-1}
$f(x) = x^3$	all reals	all reals	_____	_____
$f(x) = 2^x$	all reals	$(0, \infty)$	_____	_____
$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$	_____	_____

Row 2 explains Day 7: the exponential and the _____ are inverses, which is why their domains and ranges swapped.

Part 5 — Quick Check

- Does $f^{-1}(x)$ mean 1 divided by $f(x)$? _____
- If $(3, -5)$ is on the graph of f , name a point on f^{-1} . _____
- A graph and its inverse are reflections across which line? _____

Practice

Find each inverse. Show your work where there is room.

1. $p = \{ (0, 3), (1, 5), (2, 7) \}$ $p^{-1} = \{ \text{_____} \}$

2. Domain of $p =$ _____ Range of $p^{-1} =$ _____

3. A table gives f : $(2, 6), (4, 6), (5, 1)$. Write f^{-1} , then decide whether it is a function.

$f^{-1} = \{ \text{_____} \}$ A function? _____

4. That is a preview of Day 12. What went wrong in item 3?

5. If a graph contains $(-3, 7)$, name the matching point on its inverse. _____

Exit Ticket

A function h has the table below.

h : $(1, 4), (2, 9), (5, 0)$ $h^{-1} = \{ \text{_____} \}$

Domain of h^{-1} : _____ Range of h^{-1} : _____

Name _____ Date _____ Period _____

Writing the Inverse of a Linear Function

Algebra II | Unit 1 | Day 11 | TEK A2.2.B

Objective: I can write the inverse of a linear function algebraically by swapping x and y and solving for y , then check my answer by graphing or with a table.

Warm-Up — You Already Know How to Do This

In Algebra I you solved literal equations for a variable. That is the whole skill today.

Solve $y = 3x + 6$ for x : $x =$ _____

The Three Steps

Step 1. Replace $f(x)$ with _____.

Step 2. _____.

Step 3. Solve for y , then write it as _____.

Part 1 — Worked Examples

Example 1. $f(x) = 2x + 3$

Swap: _____ Solve: _____

So $f^{-1}(x) =$ _____

Example 2. $f(x) = 3x - 5$

Swap: _____ Solve: _____

So $f^{-1}(x) =$ _____

Part 2 — You Try

1. $f(x) = x + 7$

$f^{-1}(x) =$ _____

2. $f(x) = 4x$

$f^{-1}(x) =$ _____

3. $f(x) = 5x - 10$

$f^{-1}(x) =$ _____

4. $f(x) = -2x + 8$

$f^{-1}(x) =$ _____

Part 3 — Check It Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$ from Example 1.

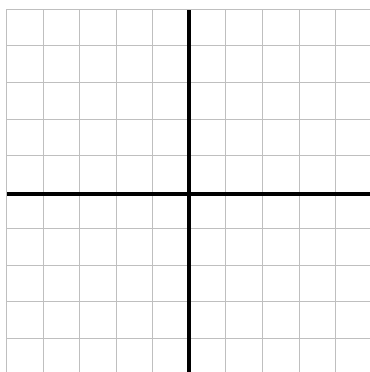
Check 1: the tables should trade columns.

x	-1	0	1	2
f(x)	1	3	5	7

x	1	3	5	7
f⁻¹(x)				

Check 2: plot both. They must mirror across $y = x$.

Graph f and f^{-1} , plus the line $y = x$.



Record it.

Slope of f : _____

Slope of f^{-1} : _____

What is the relationship?

y-intercept of f is 3. Of f^{-1} ?

Part 4 — Watch Out

A student says the inverse of $f(x) = x + 4$ is $f^{-1}(x) = 1/(x + 4)$. What went wrong?

Which linear function is its own inverse? _____

What is the inverse of a horizontal line, like $f(x) = 3$? _____

Part 5 — In Context

To convert Celsius to Fahrenheit: $F(c) = (9/5)c + 32$.

Find the inverse: $F^{-1}(x) =$ _____

What does the inverse do in words? _____

Practice

Write each inverse. Show the swap-and-solve steps.

1. $f(x) = (1/2)x - 6$

$f^{-1}(x) =$ _____

2. $f(x) = -x + 9$

$f^{-1}(x) =$ _____

Exit Ticket

Write the inverse of $f(x) = 6x - 12$, then verify it with one input.

$f^{-1}(x) =$ _____

Check: $f(3) =$ _____ and $f^{-1}(\text{_____}) =$ _____

When Is an Inverse a Function?

Algebra II | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can decide whether the inverse of a function is itself a function, and restrict the domain of the original when it is not.

Vocabulary

- Vertical line test** — if any vertical line hits the graph twice, it is _____.
- One-to-one** — every output comes from exactly _____ input. Test with a _____ line.
- Key fact** — every inverse of a function is a _____, but it is a _____ only when the original is one-to-one.
- Domain restriction** — deliberately shrinking the domain so the function _____.

Part 1 — The Problem with $f(x) = x^2$

Fill in the table, then look for repeats in the bottom row.

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$							

- Which outputs appear more than once? _____ So f is _____ one-to-one.
- Now swap to build the inverse. Both $(1, 1)$ and $(1, -1)$ appear. Is that a function? _____
- Why not? _____
- Written out, the inverse relation is $y =$ _____, and the $\pm$ is exactly the problem.

Part 2 — Fix It with a Restriction

Keep only half of the parabola, so no output repeats.

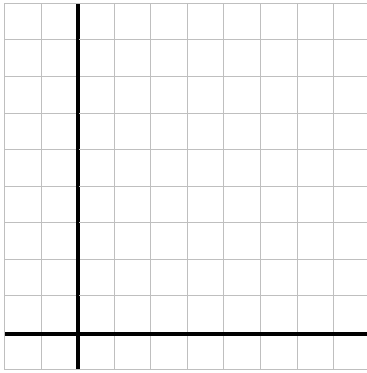
Restriction on $f(x) = x^2$	Which half of the parabola	Then $f^{-1}(x) =$
$x \geq 0$	_____	_____
$x \leq 0$	_____	_____

- Either restriction works. The usual choice is _____, which gives the familiar $f^{-1}(x) = \sqrt{x}$.
- Restricting the domain of f also restricts the _____ of f^{-1} .

Part 3 — Graph the Restricted Pair

Each square is 1 unit. Graph only the half of x^2 you kept, then its inverse, plus $y = x$.

$$f(x) = x^2 \text{ for } x \geq 0, \text{ and } f^{-1}(x) = \sqrt{x}$$



Record it.

Domain of restricted f: _____

Range of restricted f: _____

Domain of f^{-1} : _____

Do they mirror across $y = x$? _____

Part 4 — Which Pairs Need No Restriction?

Some parent functions are already one-to-one, so their inverses are functions with no help.

Function	One-to-one?	Restriction needed?	Its inverse
$f(x) = x^2$	_____	_____	_____
$f(x) = x^3$	_____	_____	_____
$f(x) = 2^x$	_____	_____	_____
$f(x) = 2x + 1$	_____	_____	_____
$f(x) = x $	_____	_____	_____

Pattern: functions that only _____ (or only decrease) are one-to-one. Ones that turn around, like x^2 and $|x|$, are not.

Part 5 — Quick Check

1. Is the inverse of every function also a function? _____
2. Which test tells you the inverse will be a function? _____
3. Why does x^3 need no restriction but x^2 does? _____

Exit Ticket

Consider $f(x) = x^2 - 4$ with no restriction.

Is its inverse a function? _____ Give a restriction that fixes it: _____

Explain in one sentence why the restriction is needed: _____

Composition to Verify Inverses

Algebra II | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions and use the results $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ to prove whether they are inverses of each other.

Vocabulary

Composition — putting one function _____ another. Written $f(g(x))$ or $(f \circ g)(x)$.

Order matters — work from the _____. Evaluate g first, then feed the result into f .

The inverse test — f and g are inverses when _____ AND _____. Both directions.

Part 1 — Warm Up on Composition

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$. Work inside first.

$$g(2) = \underline{\hspace{2cm}} \quad f(4) = \underline{\hspace{2cm}} \quad f(g(4)) = f(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$$

$$g(f(4)) = g(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}} \quad \text{Notice both came back to } \underline{\hspace{2cm}}, \text{ the number we started with.}$$

Part 2 — Prove It with Algebra

Example 1. $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$

$$f(g(x)) = 3 \cdot \underline{\hspace{2cm}} - 5 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$g(f(x)) = (\underline{\hspace{2cm}} + 5)/3 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Both give x , so f and g _____ inverses.

You Try 1. $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

You Try 2. $f(x) = x + 6$ and $g(x) = x - 6$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

Part 3 — A Pair That Is NOT Inverse

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Do not assume - compute.

$$f(g(x)) = 2(\underline{\hspace{2cm}}) + 1 = \underline{\hspace{2cm}} \quad \text{Does this equal } x? \underline{\hspace{2cm}}$$

So they are _____ inverses. What is the actual inverse of $f(x) = 2x + 1$? $f^{-1}(x) = \underline{\hspace{2cm}}$

Lesson: changing a sign is _____ the same as undoing the function. You must undo every step, in reverse order.

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{x - 3} + 4$.

Test it: is $f(f^{-1}(x)) = x$? _____ So the claim is _____.

Find the mistake. Undo the steps in reverse: subtract 4, divide by 2, take the square root, then add 3.

$$\text{Correct inverse: } f^{-1}(x) = \underline{\hspace{2cm}}$$

Why does a $\pm$ show up here? _____

What restriction would make the inverse a function? _____

Part 5 — Self-Inverse Functions

A few functions are their own inverse - composing them with themselves gives x .

1. $f(x) = x$ $f(f(x)) =$ _____ Self-inverse? _____

2. $f(x) = -x$ $f(f(x)) =$ _____ Self-inverse? _____

3. $f(x) = 1/x$ $f(f(x)) =$ _____ Self-inverse? _____

What do their graphs share? _____

Practice

Decide whether each pair is inverse. Show BOTH compositions - one direction is not proof.

1. $f(x) = 5x$ and $g(x) = x/5$

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

2. $f(x) = x - 3$ and $g(x) = 3 - x$

$f(g(x)) =$ _____ Inverses? _____ Correct inverse of f : _____

3. $f(x) = (x + 1)/2$ and $g(x) = 2x - 1$

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

4. $f(x) = x^2$ and $g(x) = \sqrt{x}$. Test with $x = -4$ as well as $x = 4$.

Does it work for every real x ? _____ Restriction needed: _____

5. Write the inverse of $f(x) = 7x + 2$, then verify it by composition.

$f^{-1}(x) =$ _____

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions.

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

Why is checking only one direction not enough? _____